Simultaneous Calibration of a Robot and a Hand-Mounted Camera

Hanqi Zhuang, *Senior Member, IEEE,* Kuanchih Wang, *Member, IEEE,* and Zvi S. Roth, *Senior Member, IEEE*

Abstract—**Visual sensing is an important feature of an intelligent robotic system. A popular configuration widely used in a variety of robotic applications is to mount a camera on the robot manipulator hand. Before performing a measurement task using such a system, both the camera and the robot need to be calibrated.**

In this paper, a procedure is developed for simultaneous calibration of a robot and a monocular camera. Unlike conventional approaches based on first calibrating the camera and then calibrating the robot, the algorithm solves for the kinematic parameters of the robot and camera in one stage, thus eliminating error propagation and improving noise sensitivity. Only two parameters are added to a robot calibration model to represent camera geometry. With this addition, different levels of calibration can be done under a unified framework. An error model relating image measurement residuals to kinematic parameter deviations is derived. Simulation and experimental studies have been conducted to assess the effectiveness of the proposed procedure.

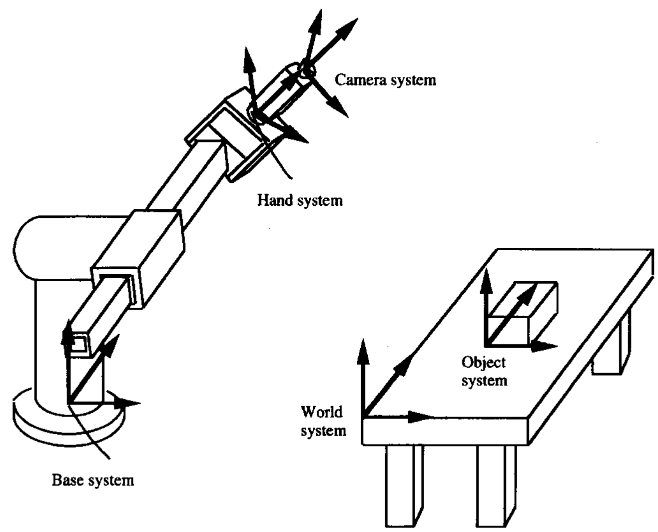


Fig. 1. A robotic system and its coordinate system assignment.

I. Introduction

VISUAL SENSING is an important feature of an intelligent robotic system. A popular configuration widely used in a variety of robotic applications is to mount a camera on the robot manipulator hand. Applications range from data acquisition within the robot work cell [1] to visual guidance for automated assembly and part transport [2].

In order to accurately measure the position and orientation of an object in a reference (world) coordinate system using the robot-camera system, various components of the system need to be calibrated. This includes the position and orientation of the robot base with respect to the robot world, the robot hand with respect to the robot base, the camera with respect to the robot hand, and the object with respect to the camera. These four tasks are respectively known as robot base calibration, manipulator calibration, hand/eye calibration and camera calibration. Fig. 1 illustrates the various coordinate systems used in this paper. In recent years, there have been extensive research efforts aimed at solving individually each of the above tasks. A user of a robot-camera system may start by calibrating the camera [3]–[6], follow by calibrating the hand/camera [7]–[11], and then use the camera and hand/eye models to calibrate the manipulator [12]–[16]. After all system components are individually calibrated, the pose of an object in the robot world system can be determined.

Such a multistage approach has pros and cons. Its main advantages are: (1) Since system calibration is performed by calibrating its components or subsystems separately, each component calibration task is relatively simple. (2) If some of the system components have changed their location or parameters, calibration needs only to be repeated for these system components. For example, if the camera changed its focal length, only the camera needs to be recalibrated.

The drawbacks of the multi-stage approach are: (1) parameter estimation errors in early stages propagate to the later stages. For example, any errors in the estimation of camera parameters may degrade the estimation quality of the robot link parameters. (2) The validity of the hand/eye calibration stage is questionable in certain cases. More specifically, most literature sources on hand/eye calibration make the assumption that the relative motions of the robot and the sensor are accurately known [6]. While the relative motion of the sensor is measured by an external device, the relative motion of the robot is computed by the use of the robot kinematic model, which may not be sufficiently accurate prior to stringent robot calibration.

An early experimental study of robot calibration using stereo cameras rigidly mounted to the robot hand is [15]. In this study, a pin-hole model was first identified for each camera

Manuscript received September 13, 1993; revised March 25, 1994.
The authors are with Robotics Center and Department of Electrical Engineering, Florida Atlantic University, Boca Raton, FL 33431 USA.
IEEE Log Number 9411923.

1042–296X/95$04.00 © 1995 IEEE

using a calibration block that consisted of planar holes and that could be moved along a direction perpendicular to the plane of the holes. Thereafter the cameras were considered to be an integral part of the robot last link. The calibration of the robot/camera system then followed using a single spherical target. In [15], the stereo cameras were calibrated using Tsai's method [5], which allows the use of a stationary precision camera calibration board. The robot/camera system was then calibrated using the same calibration board (mounted on and moved around by a coordinate measuring machine). Robot calibration measurement configurations were chosen such that the cameras were brought from about the same distance to the calibration board as that used during the camera calibration phase.

In [17], the concept of *autonomous calibration* through the creation of closed kinematic chains and use of internal sensory information only was developed. More specifically, the system set-up consisted of a stereo camera system mounted on the robot hand. Each camera was able to perform a single rotation about a selected axis. The internal sensory information included the robot joint positions, the 2-D image sensor reading and two additional joint angle readings coming from the motorized cameras. The paper proposed a simultaneous calibration strategy for both the robot and the stereo cameras using a unified mathematical model for the entire system.

In this paper, a one-stage procedure is developed for the simultaneous calibration of a robot and a monocular camera rigidly mounted to the robot hand. The work was motivated by the following sequence of logical arguments: Recognizing that a by-product of a camera calibration phase is the full pose of the camera with respect to a coordinate frame located on the camera calibration board [5], suggests a replacement of the stereo cameras with a single camera. Unlike [15], where a camera calibration step was done once to pave the way for coordinate measuring using the calibrated stereo cameras, the monocular camera needs to be recalibrated upon each change of the robot configuration, as was done in [8]. This first-step robot/camera pose measurements is to be followed by a robot kinematic identification step. The next logical reasoning is to merge these two steps using a generalized "camera model" that contains the robot kinematics.

In the proposed method, propagation errors that exist in multi-stage approaches are naturally eliminated. Unlike the technique presented in [17], which used two instrumented cameras, the new method employs a single passive camera. A single camera system has a much larger field of view, compared to a stereo cameras system, consequently enlarging the measurable workspace of the robot manipulator. In addition, the robot-camera model developed in this paper includes a radial camera lens distortion parameter that dominates other lens distortion factors [18].

With the proposed algorithm, calibration at different levels of complexity can be done under a unified framework. The paper outlines three levels of calibration. The simplest level is to identify the camera model together with the hand/eye transformation, assuming that the relative pose of the robot hand with respect to the robot base and the relative pose of the robot base with respect to the robot world are accurately known. The second level of calibration is to identify the camera model, the hand/eye transformation and the base/world transformation, assuming that the internal robot geometry is accurately known. The third level, which is the most complex, is to calibrate the camera and the entire robot simultaneously.

Among the intrinsic camera parameters, it is assumed in all three calibration levels that the image centers and the ratio of scale factors of the camera-imaging system (to be defined shortly) are given a priori. We allow the distortion coefficient and the focal length to vary in the process of robot calibration. The intrinsic camera parameters such as the image center, the ratio of scale factors and the focal length all are constant for a given camera-imaging system. They can be precalibrated once [19], [20] and then remain fixed during robot calibration. Allowing some intrinsic camera parameters to vary during the full system calibration permits measurement errors to be distributed across more parameters. This in turn may permit a better fit between the system model and the calibration data. The image center coordinates, once they are estimated, do not have significant influence on the fit between the model and the measured calibration data. On the other hand, adding the ratio of the scale factors will result in an unstable numerical system in the parameter identification stage.

One may argue that since the estimation problem is nonlinear and the dimension of the unknown parameter vector is large, such an algorithm may be very time-consuming and may require a very good initial guess. It is shown in this paper that simultaneous calibration of a camera-robot system requires the addition of only two camera parameters to the parameter set that is normally used in robot calibration. Furthermore, the method developed in the paper for the construction of error models allows robot calibration practitioners to use their own previously developed robot error models as a compatible part of the camera-robot kinematic identification software. The identification problem can thus be solved using standard nonlinear least squares procedures such as the Levenberg–Marquardt algorithm [21]. A good initial guess of the unknown parameters of the system is usually not hard to obtain since the nominal geometry of the robot is known in most cases and additional parameters can be roughly measured by proper gauging devices. It is shown that the parameter estimation algorithms typically converge in about five iterations, therefore the computation time is negligible compared to the time used for the acquiring and processing of the measurement data. On the other hand, to calibrate a robot alone, a nonlinear parameter estimation algorithm has to be used unless the robot follows special motion patterns while the pose data is collected [13], [22]–[24].

The paper is organized in the following manner. In Section II, the kinematic model of the camera-robot system is developed, the cost function for the kinematic identification is constructed, and the iterative solution algorithm is outlined. The Jacobian matrix that relates measurement errors to system parameter errors is derived in Section III. Section IV addresses a number of implementation issues. Simulation and experimental results are presented in Section V, followed by concluding remarks in Section VI.

II. Kinematic Model, Cost Function, and Solution Strategy

The basic geometry of the system is shown in Fig. 1. $\{x_w, y_w, z_w\}$ denotes the world coordinate system, normally placed at a convenient location outside the robot and camera. $\{x_b, y_b, z_b\}$ denotes the base coordinate system of the robot, physically located within the structure of the robot base. $\{x, y, z\}$ denotes the camera coordinate system, whose origin is at the optical center point O, and whose z axis coincides with the optical axis. $\{X, Y\}$ denotes the image coordinate system (not shown in Fig. 1) centered at O_I (the intersection of the optical axis z and the image plane). $\{X, Y\}$ lies on a plane parallel to the x and y axes.

Let the 4×4 homogeneous transformation $\boldsymbol{T}_n$ relating the end-effector pose to the world coordinates be

$$\boldsymbol{T}_n = \boldsymbol{A}_0 \boldsymbol{A}_1 \cdots \boldsymbol{A}_n^{-1} \boldsymbol{A}_n$$

where $\boldsymbol{A}_0$ is the 4×4 homogeneous transformation from the world coordinate system to the base coordinate system of the robot, $\boldsymbol{A}_i$ is the transformation from the $(i-1)$th to the ith link coordinate systems of the robot, and $\boldsymbol{A}_n$ is the transformation from the nth link coordinate system of the robot to the camera coordinate system.

$\boldsymbol{A}_i$ can be represented in terms of any proper kinematic modeling convention [12], [25]–[27]. Let $\boldsymbol{\rho} = [\rho_1 \quad \rho_2 \quad \cdots \quad \rho_p]^T$ be a vector consisting of all link parameters of the robot, where p is the number of independent kinematic parameters in the robot. $\boldsymbol{T}_n$ is then a matrix function of $\boldsymbol{\rho}$.

For convenience, let

$$\boldsymbol{T}_n \equiv \begin{bmatrix} R & \boldsymbol{t} \\ \boldsymbol{0}_{1\times 3} & 1 \end{bmatrix} \tag{1}$$

where

$$R \equiv \begin{bmatrix} r_1 & r_2 & r_3 \\ r_4 & r_5 & r_6 \\ r_7 & r_8 & r_9 \end{bmatrix}$$

and $\boldsymbol{t} \equiv [t_x \quad t_y \quad t_z]^T$. Clearly, r_i for $i = 1, 2, \cdots, 9$ and t_j for $j = x, y, z$ are all functions of the robot kinematic parameter vector $\boldsymbol{\rho}$.

The camera-robot model, relating the world coordinate system $\{x_w, y_w, z_w\}$ to the image coordinate system $\{X, Y\}$, is [5]

$$X[1 + \alpha(sX^2 + Y^2)] = s_x f \, \frac{r_1 x_w + r_2 y_w + r_3 z_w + t_x}{r_7 x_w + r_8 y_w + r_9 z_w + t_z} \tag{2a}$$

$$Y[1 + \alpha(sX^2 + Y^2)] = s_y f \, \frac{r_4 y_w + r_5 y_w + r_6 z_w + t_y}{r_7 x_w + r_8 y_w + r_9 z_w + t_z} \tag{2b}$$

where f is the focal length of the camera, α is the radial distortion coefficient, s_x and s_y are the camera scale factors, $s = s_y/s_x$ is the ratio of the scale factors. If s_x, s_y and f are all unknowns, one should bear in mind that these parameters are not independent. One can increase the focal length and decrease the image scale without changing the input/output relationship of the camera model.

In (2), the real image coordinates (X, Y) should be transformed to the computer image coordinates (X_f, Y_f) by

$$X = X_f - C_x$$
$$Y = Y_f - C_y$$

where X_f and Y_f denote the row and column numbers of the image pixel in the computer frame memory, and C_x and C_y are the image center coordinates.

As has been pointed in [19], it has been a common practice in the computer vision and robotics area to choose the center of the image frame buffer as the origin of the image coordinate system. It was reported in [5] that altering the image center by as much as ten pixels does not significantly influence the accuracy of 3-D measurement using the calibrated camera. Experiments on the effects of image center uncertainties to 3-D measurements were also conducted in our robotics lab, and it was observed that even when the image center coordinates are 20 pixels away from the nominal ones, the impact on the 3-D error is negligible for the purpose of calibrating the Puma 560 robot. Therefore, in this paper, the image center parameters are not included in the parameter set. If a very large image center offset is experienced in implementing camera-assisted robot calibration procedures, the image center needs to be precalibrated.

By defining

$$f_x \equiv f s_x \tag{3a}$$

$$f_y \equiv f s_y \tag{3b}$$

$$f_{xy} \equiv sX^2 + Y^2 \tag{3c}$$

(2) can be rewritten as

$$X(1 + \alpha f_{xy}) = f_x \, \frac{r_1 x_w + r_2 y_w + r_3 z_w + t_x}{r_7 x_w + r_8 y_w + r_9 z_w + t_z} \tag{4a}$$

$$Y(1 + \alpha f_{xy}) = f_y \, \frac{r_4 y_w + r_5 y_w + r_6 z_w + t_y}{r_7 x_w + r_8 y_w + r_9 z_w + t_z}. \tag{4b}$$

As seen in (1) and (4), the kinematic parameters of the robot are all absorbed into the extrinsic parameters (r_i, $i = 1, 2, \cdots, 9$) of the camera.

If the ratio of scale factors s is known, the problem of simultaneous calibration of robot and camera can be stated as follows: Given a number of calibration points whose world coordinates are known and whose image coordinates are measured, estimate the robot kinematic parameter vector $\boldsymbol{\rho}$ and the camera parameters f_x, f_y, and α. The case in which s is unknown is discussed in Section IV. In order to apply optimization techniques to solve the calibration problem, a cost function needs to be constructed.

Define a 16×1 vector $\boldsymbol{\phi}$ in the following manner:

$$\boldsymbol{\phi}(\boldsymbol{\rho}, f_x, f_y, \alpha) \equiv [f_x r_1 \quad f_y r_4 \quad r_7 \quad \alpha r_7 \quad f_x r_2 \quad f_y r_5 \quad r_8 \quad \alpha r_8 \quad f_x r_3 \quad f_y r_6 \quad r_9 \quad \alpha r_9 \quad f_x t_x \quad f_y t_y \quad t_z \quad \alpha t_z]^T \tag{5a}$$

and a 2×16 matrix C as shown in (5b) on the bottom of the page. Equation (4) can then be rewritten in the following compact form

$$C\boldsymbol{\phi}(\boldsymbol{\rho}, f_x, \alpha) = 0. \tag{6}$$

Note that the dependence of $\boldsymbol{\phi}$ on f_y is omitted as f_y can be recovered from f_x using (3).

In (6), C is a known coefficient matrix and $\boldsymbol{\phi}$ is a vector function of the unknown parameters. Given a particular calibration point whose world coordinates (x_w, y_w, z_w) and image coordinates (X, Y) are known, matrix C can be computed from (5b). According to (4), one calibration point can only provide two scalar equations. To estimate all unknown parameters, a sufficient number of calibration points have to be used. Let C_i, whose structure is given in (5b), denote the computed coefficient matrix using the ith calibration point. Let m be the number of points used for parameter estimation. The problem is then reduced to determining $\boldsymbol{\rho}$, f_x, and α that minimize the cost function E in a least squares sense, where

$$E = \sum_{i=1}^{m} [C_i\phi(\rho, f_x, \alpha)]^T [C_i\phi(\rho, f_x, \alpha)]. \tag{7}$$

An iterative procedure is needed to obtain the optimal solution. The Gauss–Newton method for solving nonlinear least square problems has the advantage of fast convergence in the neighborhood of a solution [28]. A modification of this algorithm is the more robust Levenberg–Marquardt [21] algorithm. To properly apply the Gauss–Newton algorithm to the problem at hand, the following issues have to be addressed:

1) Choice of an initial condition, and
2) The structure of the Jacobian matrix.

It is assumed for practicality that $\boldsymbol{\rho}^0$, f_x^0, and α^0, the nominal values of $\boldsymbol{\rho}$, f_x, and α are known. In the case of robot-camera calibration, the nominal robot kinematic parameter vector $\boldsymbol{\rho}^0$ is usually provided from the data sheets of the robot, except for those parameters associated with $\boldsymbol{A}_0$ and $\boldsymbol{A}_n$, which need to be roughly determined by a proper gauging device. The initial values of the scale factors s_x and s_y can be set by camera specifications. The nominal value of the distortion coefficient α is set to zero. Finally the nominal value of the focal length may be read from the lens itself.

It is assumed that the nominal parameters are in the neighborhood of the actual ones. With this assumption and from (6),

$$C_i(\boldsymbol{\phi}^0 + d\boldsymbol{\phi}) = 0 \quad i = 1, 2, \cdots, m \tag{8}$$

where the superscript 0 denotes a nominal quantity. It is further assumed that the function $\boldsymbol{\phi}$ is differentiable at the nominal parameters. Then the following relationship holds up to a first order approximation,

$$\begin{aligned} C_i\boldsymbol{\phi}^0 \cong -C_i \Bigg\{ & \frac{\partial\boldsymbol{\phi}(\rho^0, f_x^0, f_y^0, \alpha^0)}{\partial\rho} d\rho \\ & + \frac{\partial\boldsymbol{\phi}(\rho^0, f_x^0, f_y^0, \alpha^0)}{\partial\alpha} d\alpha \\ & + \Bigg[\frac{\partial\boldsymbol{\phi}(\rho^0, f_x^0, f_y^0, \alpha^0)}{\partial f_x} \\ & + s\, \frac{\partial\boldsymbol{\phi}(\rho^0, f_x^0, f_y^0, \alpha^0)}{\partial f_y} \Bigg] df_x \Bigg\} \\ & i = 1, 2, \cdots, m. \end{aligned} \tag{9}$$

The coefficient matrix in the right hand side of (9) is a Jacobian, which can either be computed explicitly as shown in the next section or be approximated using finite differences. $d\boldsymbol{\rho}$, df_x, and $d\alpha$ may be solved from (9) using a weighted least squares algorithm, and can then be used to update $\boldsymbol{\rho}$, f_x, and α. The process is iterated until the amount of adjustment on each of the parameters is below a prescribed threshold.

III. The Identification Jacobian

The Identification Jacobian is a Jacobian matrix relating measurement residuals to parameter errors $d\boldsymbol{\rho}$, df_x, and $d\alpha$. The Identification Jacobian plays a central role not only in the identification of the unknown parameter vector, but also in the study of optimal selection of measurement configurations and accuracy compensation. It is shown in this section that the original robot identification Jacobian, as derived by many robot researchers [12], [27], [30], [31], is used as one of the building blocks of the robot/camera Identification Jacobian.

It will be helpful at this point to introduce some more compact notation. A function $vec(X)$ will denote a vector valued function of a matrix X, resulting from stacking the matrix columns one on top of the other, first column on top, second column right under and so on. Thus, $\boldsymbol{\phi}$ in (5a) can be compactly rewritten as

$$\boldsymbol{\phi} = vec(\boldsymbol{F}\boldsymbol{T}_n) \tag{10}$$

where

$$\boldsymbol{F} \equiv \begin{bmatrix} f_x & 0 & 0 & 0 \\ 0 & f_y & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & \alpha & 0 \end{bmatrix}. \tag{11}$$

Assume initially that the nominal vector $\boldsymbol{\phi}^0$deviates from $\boldsymbol{\phi}^*$, the optimal solution by a small amount. Thus,

$$\boldsymbol{\phi}^* \cong \boldsymbol{\phi}^0 + d\boldsymbol{\phi} \tag{12}$$

$$C \equiv \begin{bmatrix} x_w & 0 & -x_wX & -f_{xy}x_wX & y_w & 0 & -y_wX & -f_{xy}y_wX & z_w & 0 & -z_wX & -f_{xy}z_wX & 1 & 0 & -X & -f_{xy}X \\ 0 & x_w & -x_wY & -f_{xy}x_wY & 0 & y_w & -y_wY & -f_{xy}y_wY & 0 & z_w & -z_wY & -f_{xy}z_wY & 0 & 1 & -Y & -f_{xy}Y \end{bmatrix}. \tag{5b}$$

where $d\boldsymbol{\phi}$ is a differential change of $\boldsymbol{\phi}$. To simplify the notation, the superscript 0 from now on may be omitted if no confusion arises. By (11),

$$\begin{aligned} d\boldsymbol{\phi} &\cong vec(d\boldsymbol{F}\boldsymbol{T}_n + \boldsymbol{F}\,d\boldsymbol{T}_n) \\ &= vec(d\boldsymbol{F}\boldsymbol{T}_n) + vec(\boldsymbol{F}\,d\boldsymbol{T}_n). \end{aligned} \tag{13}$$

Define the vectors

$$\begin{aligned} J_x \equiv [&r_1 \quad 0 \quad 0 \quad 0 \quad r_2 \quad 0 \quad 0 \quad 0 \\ &r_3 \quad 0 \quad 0 \quad 0 \quad t_x \quad 0 \quad 0 \quad 0]^T \end{aligned} \tag{14a}$$

$$\begin{aligned} J_y \equiv [&0 \quad r_4 \quad 0 \quad 0 \quad 0 \quad r_5 \quad 0 \quad 0 \\ &0 \quad r_6 \quad 0 \quad 0 \quad 0 \quad t_y \quad 0 \quad 0]^T \end{aligned} \tag{14b}$$

$$\begin{aligned} J_\alpha \equiv [&0 \quad 0 \quad 0 \quad r_7 \quad 0 \quad 0 \quad 0 \quad r_8 \\ &0 \quad 0 \quad 0 \quad r_9 \quad 0 \quad 0 \quad 0 \quad t_z]^T. \end{aligned} \tag{14c}$$

By simple algebraic manipulations, the first term in the right hand side of (13), representing the contribution to the error model by the camera intrinsic error parameters, can then be written as

$$vec(d\boldsymbol{F}\boldsymbol{T}_n) = (J_x + sJ_y)\,df_x + J_\alpha\,d\alpha. \tag{15}$$

More involved algebraic manipulations need to be performed to expand the second term in the right hand side of (13), representing the contribution to the error model by the robot kinematic error parameters. A main objective is to allow robot calibration researchers and practitioners to retain the original robot error model, hence a large portion of the kinematic identification software. Define

$$\Delta \equiv \boldsymbol{T}_n^{-1}\,d\boldsymbol{T}_n. \tag{16}$$

Δ, a 4×4 matrix, has the following structure [25],

$$\Delta = \begin{bmatrix} \Omega(\delta) & \boldsymbol{d} \\ \boldsymbol{0}_{1\times 3} & 0 \end{bmatrix} \tag{17}$$

where $\boldsymbol{\delta} \equiv [\delta_x,\, \delta_y,\, \delta_z]^T$ and $\boldsymbol{d} \equiv [d_x,\, d_y,\, d_z]^T$ are respectively the 3×1 rotational and positional error vectors of $\boldsymbol{T}_n$, and $\Omega(\delta)$ is the following skew-symmetric matrix,

$$\Omega(\delta) = \begin{bmatrix} 0 & \delta_z & -\delta_y \\ -\delta_z & 0 & \delta_x \\ \delta_y & -\delta_x & 0 \end{bmatrix}.$$

The 6×1 vector $[\boldsymbol{d}^T,\, \boldsymbol{\delta}^T]^T$ is said to be the *pose error vector of* $\boldsymbol{T}_n$.

Two steps are now needed to relate $vec(\boldsymbol{F}\,d\boldsymbol{T}_n)$ to the robot kinematic parameter error vector $d\boldsymbol{\rho}$. First, $vec(\boldsymbol{F}\,d\boldsymbol{T}_n)$ is related to the pose error vector $[\boldsymbol{d}^T,\, \boldsymbol{\delta}^T]^T$ by a linear transformation. $vec(\boldsymbol{F}\,d\boldsymbol{T}_n)$ is then related to $d\boldsymbol{\rho}$ by using an additional linear transformation mapping $d\boldsymbol{\rho}$ to $[\boldsymbol{d}^T,\, \boldsymbol{\delta}^T]^T$.

Multiplying both sides of (16) by $\boldsymbol{F}\boldsymbol{T}_n$ yields

$$\boldsymbol{F}\,d\boldsymbol{T}_n = \boldsymbol{F}\boldsymbol{T}_n\Delta. \tag{18}$$

$\boldsymbol{F}$ can be decomposed into

$$\boldsymbol{F} \equiv \boldsymbol{G}\boldsymbol{H}$$

where

$$\boldsymbol{G} \equiv \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & \alpha & 0 \end{bmatrix} \tag{19a}$$

$$\boldsymbol{H} \equiv \begin{bmatrix} f_x & 0 & 0 & 0 \\ 0 & f_x & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}. \tag{19b}$$

Then from (18),

$$vec(\boldsymbol{F}\,d\boldsymbol{T}_n) = vec(\boldsymbol{F}\boldsymbol{T}_n\Delta) = vec(\boldsymbol{G}\boldsymbol{H}\boldsymbol{T}_n\Delta). \tag{20a}$$

A crucial step in the derivation of the robot/camera error model is to invoke the Kronecker product of matrices. That is,

$$\begin{aligned} vec(\boldsymbol{G}\boldsymbol{H}\boldsymbol{T}_n\Delta) &= (\boldsymbol{I} \otimes \boldsymbol{G})\,vec(\boldsymbol{H}\boldsymbol{T}_n\Delta) \\ &= (\boldsymbol{I} \times \boldsymbol{G}) \begin{bmatrix} vec\begin{pmatrix} \boldsymbol{H}_{3:3}R\,\Omega(\delta) \\ \boldsymbol{0}_{1\times 3} \end{pmatrix} \\ \boldsymbol{H}_{3:3}R\boldsymbol{d} \\ 0 \end{bmatrix} \end{aligned} \tag{20b}$$

where $\boldsymbol{I}$ is a 3×3 identity matrix, $\boldsymbol{H}_{3:3}$ is the upper-left 3×3 submatrix of $\boldsymbol{H}$, and R and $\boldsymbol{d}$ are the rotation matrix and translation vector of $\boldsymbol{T}_n$. The symbol $\otimes$ denotes a Kronecker product of two matrices [29]; that is,

$$A \otimes B \equiv \begin{bmatrix} a_{11}B & \cdots & a_{1n}B \\ \vdots & & \vdots \\ a_{n1}B & \cdots & a_{nn}B \end{bmatrix}$$

where a_{ij} is the ijth elements of A.

In the error-model based robot calibration literature, the transformation from $d\boldsymbol{\rho}$ to $[\boldsymbol{d}^T,\, \boldsymbol{\delta}^T]^T$ is available. More specifically, one has

$$\boldsymbol{\delta} = \sum_{j=1}^{p} J_{\delta j}\,d\rho_j \tag{21a}$$

$$\boldsymbol{d} = \sum_{j=1}^{p} J_{dj}\,d\rho_j \tag{21b}$$

where $J_{\delta i}$ and J_{di} are 3×1 vectors. The details of $J_{\delta i}$ and J_{di} in terms of a specific kinematic modeling convention can again be found in [12], [26], [29], [30]. After substituting (21) into (20b) and with some algebraic manipulations, one obtains

$$\begin{aligned} \mathrm{vec}(\boldsymbol{F}\,d\boldsymbol{T}_n) = (\boldsymbol{I} \otimes \boldsymbol{G}) \sum_{j=1}^{p} \\ \cdot \begin{bmatrix} vec\begin{pmatrix} \boldsymbol{H}_{3:3}R\Omega(J_{\delta j}) \\ \boldsymbol{0}_{1\times 3} \end{pmatrix} \\ \boldsymbol{H}_{3:3}RJ_{dj} \\ 0 \end{bmatrix} d\rho_j. \end{aligned} \tag{22}$$

Substituting (15) and (22) into (8) yields

$$\begin{aligned} C_i\boldsymbol{\phi} \cong\ & -C_iJ_\alpha\,d\alpha - C_i(J_x + sJ_y)\,df_x \\ & - C_i(\boldsymbol{I} \otimes \boldsymbol{G}) \sum_{j=1}^{p} \begin{bmatrix} vec\begin{pmatrix} \boldsymbol{H}_{3:3}R\Omega(J_{\delta j}) \\ \boldsymbol{0}_{1\times 3} \end{pmatrix} \\ \boldsymbol{H}_{3:3}RJ_{dj} \\ 0 \end{bmatrix} d\rho_j \\ & i = 1,\, 2,\, \cdots,\, m. \end{aligned} \tag{23}$$

This equation is now in the form of (9). To write the result in a matrix form, let

$$J_i \equiv \Bigg\{ -C_i J_\alpha \quad -C_i(J_x + sJ_y) \quad C_i(\boldsymbol{I} \otimes \boldsymbol{G}) \cdot \begin{bmatrix} vec\begin{pmatrix} \boldsymbol{H}_{3:3} R\Omega(J_{\delta 1}) \\ \boldsymbol{0}_{1\times 3} \end{pmatrix} \\ \boldsymbol{H}_{3:3} R J_{d1} \\ 0 \end{bmatrix} \cdots C_i(\boldsymbol{I} \otimes \boldsymbol{G}) \begin{bmatrix} vec\begin{pmatrix} \boldsymbol{H}_{3:3} R\Omega(J_{\delta p}) \\ \boldsymbol{0}_{1\times 3} \end{pmatrix} \\ \boldsymbol{H}_{3:3} R J_{dp} \\ 0 \end{bmatrix} \Bigg\}. \tag{24}$$

J_i is a $2 \times (p+2)$ matrix; and also let

$$d\boldsymbol{\rho}^{aug} \equiv [d\alpha \quad df_x \quad d\boldsymbol{\rho}^T]^T. \tag{25}$$

$d\boldsymbol{\rho}^{aug}$ is a $(p+2) \times 1$ vector. Equation (23) can then be rewritten as

$$C_i \boldsymbol{\phi} = J_i \, d\boldsymbol{\rho}^{aug} \quad i = 1, 2, \cdots, m. \tag{26}$$

Let

$$\boldsymbol{C} \equiv \begin{bmatrix} C_1 \\ C_2 \\ \vdots \\ C_m \end{bmatrix} \tag{27a}$$

and

$$\boldsymbol{J} \equiv \begin{bmatrix} J_1 \\ J_2 \\ \vdots \\ J_m \end{bmatrix}. \tag{27b}$$

Equation (26) can then be written in a single matrix form,

$$\boldsymbol{C\phi} = \boldsymbol{J} \, d\boldsymbol{\rho}^{aug}. \tag{28}$$

Equation (28) provides the relationship between the measurement residual error vector $\boldsymbol{C\phi}$ and the augmented parameter error vector $d\boldsymbol{\rho}^{aug}$. $\boldsymbol{J}$ is termed the *Identification Jacobian* for robot-camera calibration.

IV. Implementation Issues

A. Camera Parameters

The distortion parameter α needs not be computed each time the robot and the camera systems are calibrated. If α is known, the error model derived in the last section can be simplified. In this case, $\boldsymbol{\phi}$ is a 12×1 vector,

$$\boldsymbol{\phi} \equiv [f_x r_1 \quad f_y r_4 \quad r_7 \quad f_x r_2 \quad f_y r_5 \quad r_8 \quad f_x r_3 \quad f_y r_6 \quad r_9 \quad f_x t_x \quad f_y t_y \quad t_z]^T \tag{29a}$$

C_i is a 2×12 matrix, (see (29b) below) where the subscript i is omitted. J_x and J_y are 12×1 vectors,

$$J_x \equiv [r_1 \quad 0 \quad 0 \quad r_2 \quad 0 \quad 0 \quad r_3 \quad 0 \quad 0 \quad t_x \quad 0 \quad 0]^T \tag{29c}$$
$$J_y \equiv [0 \quad r_4 \quad 0 \quad 0 \quad r_5 \quad 0 \quad 0 \quad r_6 \quad 0 \quad 0 \quad t_y \quad 0]^T \tag{29d}$$

J_i is a $2 \times (p+2)$ matrix,

$$J_i \equiv \Bigg\{ -C_i J_x \quad -C_i J_y \quad C_i \begin{bmatrix} [\boldsymbol{I} \otimes (\boldsymbol{H}_{3:3} R)] \, vec[\Omega(J_{\delta 1})] \\ \boldsymbol{H}_{3:3} R J_{d1} \end{bmatrix} \cdots C_i \begin{bmatrix} [\boldsymbol{I} \otimes (\boldsymbol{H}_{3:3} R)] \, vec[\Omega(J_{\delta p})] \\ \boldsymbol{H}_{3:3} R J_{dp} \end{bmatrix} \Bigg\} \tag{29e}$$

and $d\boldsymbol{\rho}^{aug}$ is a $(p+2) \times 1$ vector,

$$d\boldsymbol{\rho}^{aug} \equiv [df_x, \, df_y, \, d\boldsymbol{\rho}^T]^T. \tag{29f}$$

The above new set of equations should be used to replace (5), (14), and (25) for parameter identification if α is known.

The scale factors s_x and s_y of the camera can be determined using camera calibration techniques such as that given in [19] or be identified together with the robot parameters and the focal length by the approach given in this paper. These parameters need only to be determined once for a particular camera system. For the system given in (29), the scale factors do not need to be known in advance as these are identified as combinations with other parameters.

B. Robot Parameters

A sufficient number of independent link parameters need to be used to express any variation of the actual robot structure away from the nominal design. This number, for a serial manipulator consisting of rigid links connected by low pair joints, is $4N - 2P + 6$, where N is the number of degrees of freedom and P is the number of prismatic joints [12]. For example, in a PUMA 560 robot, the number of link parameters, hence the dimension of the PUMA kinematic parameter vector $\boldsymbol{\rho}$ is 30.

Not all parameters in $\boldsymbol{\rho}$ need to be calibrated each time. It is natural to accommodate different complexities of calibration with the approach proposed above. We discuss three levels of calibration next. Readers are referred to [27] for details of $J_{\delta i}$ and J_{di} used in (21) which can be applied to each level of calibration.

As has been mentioned, the simplest level of calibration is to identify the camera parameters α, f_x, and f_y together with the parameters that specify the hand/eye transformation $\boldsymbol{A}_n$, assuming that the transformation from the world coordinate system to the robot hand coordinate system is known. This type of calibration is necessary whenever the relative pose of the camera with the robot is changed. We use six link parameters to specify the transformation $\boldsymbol{A}_n$. Together with the two camera parameters, there are eight parameters to be estimated. That is, the dimension of $d\boldsymbol{\rho}^{aug}$ given in (25) is eight.

$$C \equiv \begin{bmatrix} x_w & 0 & -x_w X & y_w & 0 & -y_w X & z_w & 0 & -z_w X & 1 & 0 & -X \\ 0 & x_w & -x_w Y & 0 & y_w & -y_w Y & 0 & z_w & -z_w Y & 0 & 1 & -Y \end{bmatrix} \tag{29b}$$

The second level of calibration is to identify the camera parameters together with the parameters that specify the hand/eye transformation $\boldsymbol{A}_n$ and the base/world transformation $\boldsymbol{A}_0$, assuming that the robot geometry is accurately known. This type of calibration is necessary whenever the camera changes its location with respect to the robot hand and the robot also changes its location with respect to an external reference object. Since four additional parameters in $\boldsymbol{A}_0$ needs to be identified, the dimension of $\boldsymbol{\rho}^{aug}$ is twelve.

The third level, which is the most complex level of calibration, is to calibrate the entire robot-camera system. In this case, the dimension of $\boldsymbol{\rho}^{aug}$ is $4N - 2P + 8$, among which two are camera parameters.

C. Change of Reference Frame

$[\boldsymbol{d}^T, \boldsymbol{\delta}^T]^T$, the pose error vector of $\boldsymbol{T}_n$ defined in Section III, is represented in the world coordinate system since $\boldsymbol{T}_n$ is defined as the transformation from the world coordinate system to the camera coordinate system. The robot error model given in (21) should be consistent with this convention. If the pose error vector is represented in the camera coordinate system, as has been the case in some robot kinematic error models, the expression given in (24) has to be modified to accommodate the change of the reference coordinate system.

The transformation from the camera coordinate system to the world coordinate system is $\boldsymbol{T}_n^{-1}$. Let $\boldsymbol{d}'$ and $\boldsymbol{\delta}'$ be respectively the rotational and positional error vectors of $\boldsymbol{T}_n^{-1}$. Also let

$$\boldsymbol{T}_n^{-1} \equiv \begin{bmatrix} R' & \boldsymbol{t}' \\ \boldsymbol{0}_{1\times 3} & 1 \end{bmatrix}$$

Then

$$R = (R')^T \tag{30a}$$
$$\boldsymbol{t} = -(R')^T \boldsymbol{t}'. \tag{30b}$$

Perturbing both sides of (30) yields

$$dR = (dR')^T \tag{31a}$$
$$\boldsymbol{d} = -(dR')^T \boldsymbol{t}' - (R')^T \boldsymbol{d}' \tag{31b}$$

where by definition $\boldsymbol{d} = d\boldsymbol{t}$ and $\boldsymbol{d}' = d\boldsymbol{t}'$. From (31a), one has

$$\Omega(\boldsymbol{\delta}) = -R^T \Omega(\boldsymbol{\delta}') R \tag{32}$$

which, after simple vector and matrix manipulations, reduces to

$$\boldsymbol{\delta} = -R^T \boldsymbol{\delta}'. \tag{33}$$

From (31b),

$$\begin{aligned} \boldsymbol{d} &= -dR\boldsymbol{t}' - R\boldsymbol{d}' \\ &= R\Omega(\boldsymbol{\delta}')\boldsymbol{t}' - R\boldsymbol{d}' \end{aligned}$$

or

$$\boldsymbol{d} = -R\Omega(\boldsymbol{t}')\boldsymbol{\delta}' - R\boldsymbol{d}'. \tag{34}$$

$\boldsymbol{\delta}$ and $\boldsymbol{d}$ in (20b) shall be substituted by (33) and (34) whenever the robot error model (i.e., $J_{\delta i}$ and J_{di}) is given in terms of the camera coordinate frame. Equations (21)–(24) shall also be modified accordingly.

D. Observability of the Unknown Parameters

In the robot calibration literature, the observability of the kinematic error parameter vector $d\boldsymbol{\rho}$ is defined in terms of the Identification Jacobian. If the Identification Jacobian is full rank, the error parameter vector is said to be observable.

It is difficult to obtain analytical observability results since the structure of the Jacobian matrix in this case is very complex [refer to (3.16a)]. However, the Identification Jacobian derived in this paper can be used for optimal off-line search of robot measurement configurations, which can significantly improve calibration quality. Borm and Menq [33] defined observability measure for robot calibration using all singular values of the Identification Jacobian. As observed experimentally, a low number of well-selected measurements can produce superior results, compared with identification done based on a set of large number of randomly selected measurement configurations. Similar observations were made by Driels and Pathre [34] using the condition number of the Identification Jacobian. The extension of the techniques to robot-camera calibration is straightforward after the identification Jacobian matrix is made available.

E. Verification of the Calibration Results

It is difficult to obtain highly accurate reference against which the accuracy performance of a robot-camera calibration task is evaluated. In the absence of an accurate external device, one may use the following two approaches to assess the accuracy performance.

Approach I—Verification on the image plane: Assume that f_x, f_y, α, and $\boldsymbol{\rho}$ have been identified. A set of robot configurations, which are different from those used for identification, is given. A set of world coordinates of the calibration points in each robot configuration is also given. The coordinates of the corresponding image points are measured. Using the identified camera and robot parameters together with the robot joint variables at each configuration, and the world coordinates of the calibration points, one can compute the predicted image coordinates of each image point [refer to (2)]. The Euclidean norm of the difference between the measured and computed image coordinates at each image point can be defined as a *2-D calibration error.* After 2-D calibration errors are computed for all image points at all robot configurations, the mean and standard deviation of the 2-D calibration errors can be obtained.

Approach II—Verification in the world coordinate system: One may compute 3-D world coordinates of each calibration point using the calibrated robot and camera parameters. This is possible by using more than one view, that is more than one robot configuration, for the same calibration point. Two views of a point are sufficient to compute its world coordinates using stereo triangulation. Using more than two views calls for a least squares fitting. The Euclidean norm of the difference between the computed world coordinates of the calibration point and its given world coordinates can be defined as a *3-D calibration error.* One can compute the mean and standard deviation of the 3-D calibration errors by repeating this procedure for all calibration points.

TABLE I
NOMINAL MCPC LINK PARAMETERS FOR THE PUMA 560

Link #	x_i (mm)	α_i (deg)	y_i (mm)	β_i (deg)	z_6 (mm)	γ_6 (deg)
0	-100	90	-1000	180		
1	0	-90	-260	0		
2	431.8	0	0	0		
3	-20.3	90	149.09	0		
4	0	-90	-433.07	0		
5	0	90	0	0		
6	0	0	0	0	300	0

V. SIMULATION AND EXPERIMENTAL RESULTS

A. Simulation Results

The simulation studies were performed using the geometry of the Puma 560 robot. A modified complete and parametrically continuous (MCPC) model [27] was adopted to construct the robot kinematic model and error model, although other kinematic modeling conventions suitable for calibration can also be used for this purpose. The nominal parameters of the Puma 560 are given in Table I.

In the MCPC model, the origin of frame $\{i\}$ is placed on the intersection point of joint axis $i+1$ on a plane perpendicular to axis $i+1$ which passes through the origin of frame $\{i-1\}$, except for the world and the last link frames. The z_i axis of each frame is pointed in the direction of joint axis $i+1$. Each link transformation A_i is given as [27]:

$$\begin{aligned} A_i &= Rot(z, \theta_i)Rot(x, \alpha_i)Rot(y, \beta_i) \\ &\quad \cdot Trans(x_i, y_i, 0) \quad i = 0, 1, \cdots, 5 \\ A_n &= Rot(z, \theta_n)Rot(x, \alpha_n)Rot(y, \beta_n) \\ &\quad \cdot Rot(z, \gamma_n)Trans(x_n, y_n, z_n) \end{aligned}$$

where $\theta_0 = 0$ and θ_i, for $i = 1, 2, \cdots, 6$, are the joint variables. In this convention, the joint variables are not part of the kinematic parameter set to be identified.

To simulate an actual robot, randomly selected parameter deviations uniformly distributed at ± 2 mm for translational parameters and ± 0.005 rad for orientational parameters were added to the robot kinematic parameters. The following three issues were investigated:

1) Number of iterations required for the algorithm to converge.
2) Accuracy performance of the proposed algorithm under different levels of noise intensities.
3) Accuracy performance comparison between the proposed one-stage algorithm and a two-stage algorithm based on recalibration of the camera at every robot calibration measurement configuration. For the two-stage algorithm, Tsai's camera calibration technique was used to compute robot poses at those measurement configurations. The computed poses were then used to identify the robot geometry.

Table II lists three different types of noise processes used in simulations. $U[a, b]$ denotes a uniformly distributed random variable in the interval $[a, b]$. Type I noise models a moderate noise level for both position and orientation measurements. Type II has a larger orientation noise level. Type III models a degraded measurement system.

The Levenberg–Marquardt nonlinear least-squares algorithm was applied to identify link parameters under different conditions. The initial condition for the robot link parameters was listed in Table I. It was experienced that for all tries, the algorithm converged within five iterations. The accuracy performance of the algorithms was evaluated using the 3-D approach described in the previous section. Representative simulation results are shown in Figs. 2–4. The horizontal axes in these figures represent the number of homogeneous transformation equations (note that each consists of two algebraic equations) for the estimation problem. The vertical axes of these figures provide the mean and standard deviation of the rotation and position estimation errors. For the one-stage algorithm, five image points were measured in each robot configuration. Similarly, five object points were measured to determine a pose in the two-stage algorithm.

By studying the simulation results, the following remarks are made:

1) As expected, the estimation error increases roughly proportionally to the noise intensity (comparing Figs. 2–4).
2) The one-stage algorithm generally outperforms the two-stage algorithm in terms of accuracy (Fig. 4).

TABLE II
NOISE INTENSITY LEVELS USED IN SIMULATION

	Position noise (mm)	Orientation noise (rad)
Type I	U[-0.1, 0.1]	U[-0.0001, 0.0001]
Type II	U[-0.1, 0.1]	U[-0.0005, 0.0005]
Type III	U[-0.5, 0.5]	U[-0.0005, 0.0005]

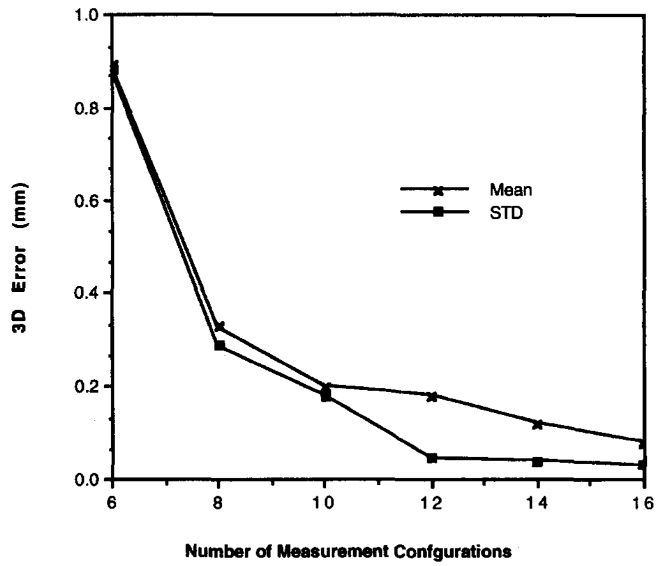


Fig. 2. Performance of one-stage approach under Type-I noise.

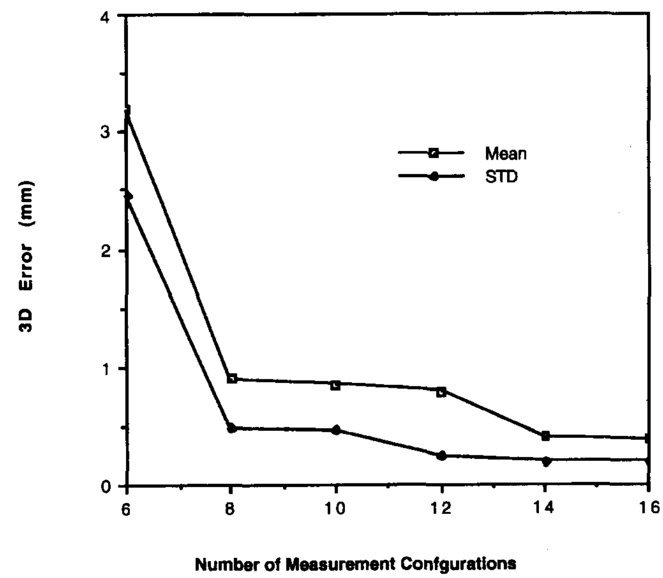


Fig. 3. Performance of one-stage approach under Type-III noise.

3) By using additional measurements, estimation errors decrease gradually until the number of measurements is greater than ten, thereafter no significant improvement has been observed.

B. Experimental Results

The experiment system consisted of a Puma 560 robot, a CCD camera (Electrophysics, model number CCD1200), a 486 personal computer, an ITEX video imaging board with its driver software, a camera calibration board, and a coordinate measuring machine (CMM) (refer to Fig. 5). The CMM was used to move the calibration board to different locations.

The CCD camera has 510H by 492V picture elements with 8.8×6.6 mm^2 sensing area. The camera calibration board is a customized glass plate painted with vapor deposited metallic chromium. It has 10×10 dot array points with center to center distance of 10 mm, and diameter (of each point) of 2 mm. The flatness of the calibration board is within ± 0.003 mm and the center to center accuracy of the calibration points is within ± 0.002 mm.

The vision algorithms for camera calibration were written in Microsoft-C. To estimate the image coordinates of a calibration point, an adaptive thresholding algorithm was devised. The

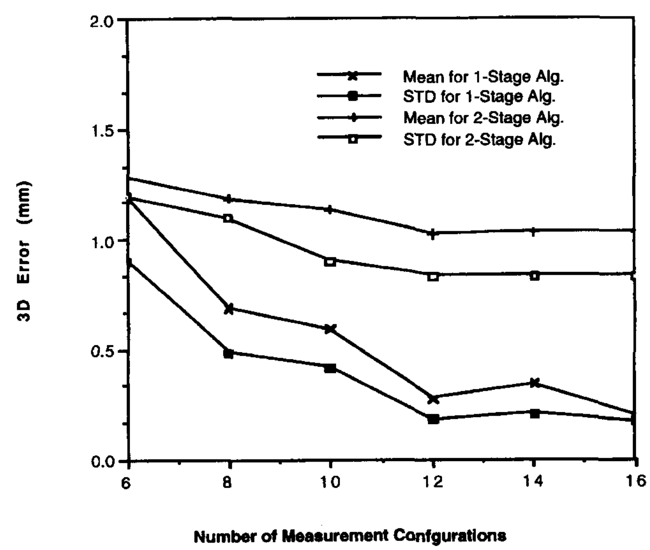


Fig. 4. Comparison of one-stage and two-stage approaches.

algorithm first smoothed an image of the camera calibration board by applying a 3×3 low-pass mask three times. It then detected consecutively each calibration point, and computed the histogram within a window surrounding each point. The

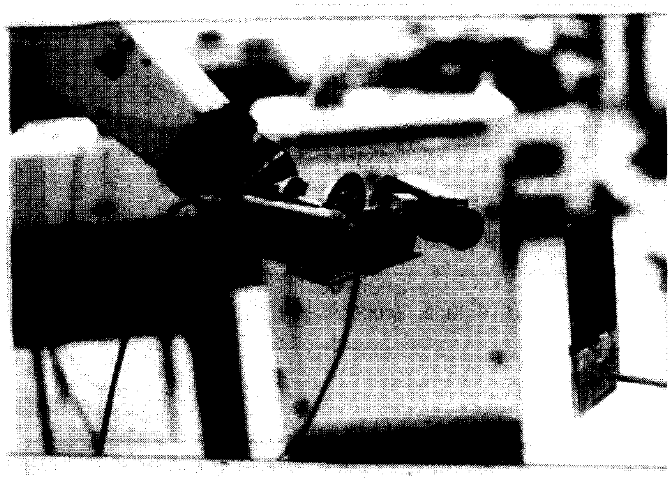


Fig. 5. The experiment system setup.

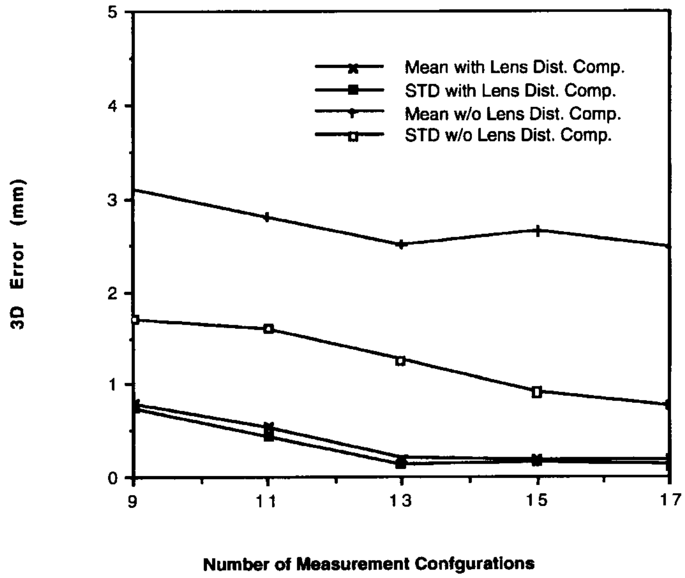


Fig. 6. Performance comparison for system calibration with or without lens distortion compensations.

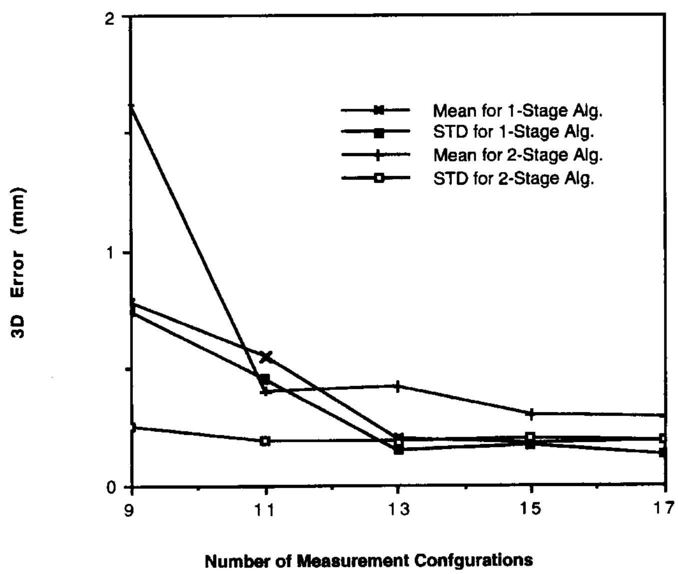


Fig. 7. Comparison of one-stage and two-stage approaches.

intersection of two Gaussian curves which fit the histogram was chosen as the threshold value for the calibration point. The centroid of the image of calibration point was selected as the estimate of its image coordinates. This procedure could yield image coordinates accuracy to within 1/5 of a pixel.

The following accuracy issues were studied experimentally:

1) The performance of the one-stage algorithm with and without taking into account lens distortion.
2) Performance comparison between the one-stage algorithm and a two-stage algorithm.
3) The performance of the one-stage algorithm with various modeling complexities.

Note that in studies 2 and 3, the one-stage algorithm with lens distortion was used. Furthermore, in item 3, it is assumed that to calibrate a particular subsystem of the robotic system, the remaining parts have been precalibrated.

The initial conditions of the robot link parameters for the Levenberg–Marquardt algorithm are listed in Table I with the exception that the initial condition for *BASE*, the transformation from the robot base frame to the world frame, and *TOOL*, the transformation from the camera frame to the robot tool frame, were provided by crude measurements.

Four data sets with a total of 52 measurement configurations were recorded. One data set out of the four was used for the purpose of verification. At each measurement configuration, an average number of five calibration points on the calibration board and the corresponding image points were actually used in parameter identification. Transformation between the CMM and the calibration board, which was mounted on the CMM probe, can be obtained by separately moving the calibration board in the x, y, and z directions of the CMM. Thus the world coordinates of the calibration dots on the calibration board can be computed after each CMM movement. The accuracy performance of the algorithms was again evaluated using 3-D errors. Figs. 6–8 illustrates the experimental results.

The following observations can be made based on the experimental results:

1) The accuracy performance of the one-stage algorithm is very promising. The error residual is at the level of 0.2 mm following the robotic system calibration (refer to Fig. 6).
2) Inclusion of the lens distortion coefficient in the robot-camera model significantly improves the accuracy performance of the one stage algorithm (refer to Fig. 6).
3) Experimental results confirm that the one-stage algorithm in general outperforms the two-stage algorithm (refer to Fig. 7). Note that the accuracy performance of the two-stage algorithm depends highly on the that of the camera calibration procedure. In our case, the 3-D error of the camera model is about 0.05 mm.
4) If the rest of the system has been calibrated in advance, partial calibration can achieve an accuracy compatible with that of entire robotic system calibration (refer to Fig. 8).

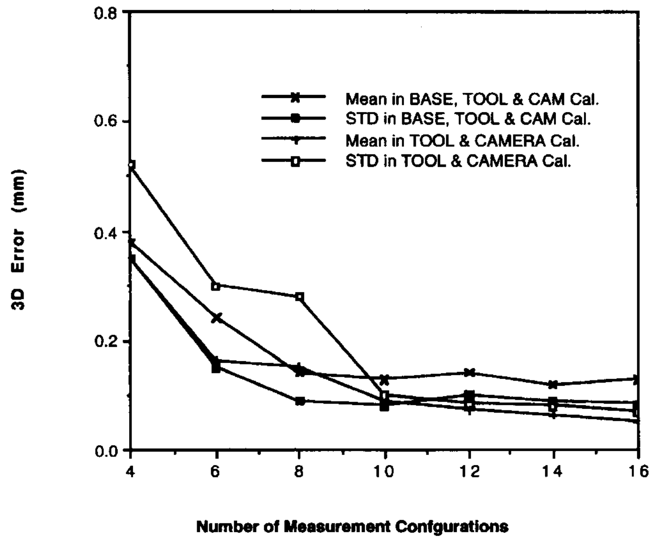


Fig. 8. Results of BASE, TOOL & CAMERA calibration and TOOL & CAMERA calibration.

VI. Concluding Remarks

Calibration is at the very heart of creating a truly off-line robotic programming environment. Autonomous calibration is of particular importance for any robot that must function outside a controlled laboratory environment [18]. This paper has presented an autonomous calibration method based on a single hand-held camera. The proposed method has the following features:

1) The camera and robot parameters are identified simultaneously. Thus propagation errors which exist in multi-stage calibration approaches are eliminated. On the other hand, accuracy performance of the two-stage algorithm depends highly on the accuracy of the camera calibration procedure. All this is reflected in the experimental results.
2) By calibrating the radial lens distortion coefficient, the accuracy performance of this procedure can be greatly improved. This is true for both camera calibration and for camera-assisted robot calibration. The proposed method allows the user to calibrate the distortion parameter together with robot parameters.
3) The calibration scheme uses a monocular camera, by which the field-of-view of the camera can be larger than that of a pair of stereo cameras. Moreover, the processing speed of one camera system is faster.
4) Only two camera parameters were added to the robot calibration model. With this addition, different levels of calibration can be done under a unified framework.
5) By employing the Levenberg–Marquardt algorithm, the calibration procedure converges rapidly, as a good set of initial conditions for the robot and camera parameters is usually available.
6) The approach can be automated if the calibration fixture is designed such that the robot can be moved to a sufficient large portion of its feasible work space yet can still view the calibration fixture with good resolution. A research in under way to fully automate the calibration process.

The major limitations of the proposed approach are:

1) Since only one camera is used, the system cannot recover the absolute distance information of the scene, unless robot motions are introduced to emulate a stereo-camera system.
2) Like any vision-based robot calibration scheme, if the camera position with respect to the robot hand is changed, the transformation from the camera to the robot hand has to be recalibrated.
3) The extrinsic parameters of the camera are absorbed into the robot parameters. This is equivalent to treating the camera coordinate system as an "imaginary tool system". If an actual tool is also attached to the robot hand, the transformation from the tool to the camera has to be determined separately. This problem exists in all robotic systems equipped with a hand/eye configuration.

Acknowledgment

The authors would like to express sincere gratitude toward the reviewers for their valuable suggestions and comments.

References

[1] T. Stahs and F. Wahl, "Fast and versatile range data acquisition in a robot work cell," in *Proc. IEEEI/RSJ Int. Conf. Intell. Robots Syst.*, Raleigh, NC, July 1992, pp. 1169–1174.
[2] D. Vernon and M. Tistarelli, "Using camera motion to estimate range for robotic parts manipulation," *IEEE Trans. Robot. Automat.*, vol. 6, pp. 509–521, Oct. 1990.
[3] H. A. Martins, J. R. Birk, and R. B. Kelley, "Camera models based on data from two calibration planes," *Computer Graphics and Image Process.*, vol. 17, pp. 173–180, 1981.
[4] I. Sutherland, "Three-dimensional data input by tablet," *Proc. IEEE*, vol. 62, pp. 453–461, Apr. 1974.
[5] R. Y. Tsai, "A versatile camera calibration technique for high accuracy 3-D machine vision metrology using off-the-shelf TV cameras and lenses," *IEEE J. Robot. Automat.*, vol. RA-3, pp. 323–344, Aug. 1987.
[6] Y. Liu, T. S. Huang, and O. D. Faugeras, "Determination of camera location from 2-D to 3-D line and point correspondences," *IEEE Trans. Patt. Anal. Mach. Intell.*, vol. 12, pp. 13–27, Jan. 1990.
[7] Y. C. Shiu and S. Ahmad, "Calibration of wrist-mounted robotic sensors by solving homogeneous transform equations of the form $AX = XB$," *IEEE Trans. Robot. Automat.*, vol. 5, pp. 16–27, Feb. 1989.
[8] R. Y. Tsai and R. K. Lenz, "A new technique for fully autonomous and efficient 3-D robotics hand/eye calibration," *IEEE Trans. Robot. Automat.*, vol. 5, pp. 345–357, June 1989.
[9] J. K. Chou and M. Kamel, "Finding the position and orientation of a sensor on a robot manipulator using quaternions," *Int. J. Robot. Res.*, vol. 10, no. 3, pp. 240–254, June 1991.
[10] H. Zhuang and Z. S. Roth, "A note on 'Calibration of wrist-mounted robotic sensors by solving homogeneous transform equations of the form $AX = XB$',"' *IEEE Trans. Robot. Automat.*, vol. 7, pp. 877–878, Dec. 1991.
[11] H. Zhuang and Y. Shiu, "A noise-insensitive algorithm for robotic hand/eye calibration with or without sensor orientation measurement," *IEEE Trans. Syst. Man Cybernet.*, vol. 23, pp. 1168–1175, 1993.
[12] B. M. Mooring, Z. S. Roth, and M. R. Driels, *Fundamentals of Manipulator Calibration.* New York: Wiley, 1991.
[13] R. K. Lenz and R. Y. Tsai, "Calibrating a Cartesian robot with eye-on-hand configuration independent of eye-to-hand relationship," *IEEE Trans. Patt. Anal. Mach. Intell.*, vol. 11, pp. 916–928, Sept. 1989.
[14] M. R. Driels and U. S. Pathre, "Vision-based automatic theodolite for robot calibration," *IEEE Trans. Robot. Automat.*, vol. 7, pp. 351–360, June 1991.
[15] H. Zhuang, Z. S. Roth, and K. Wang, "Robot calibration by mobile camera system," in *Proc. ASME Winter Ann. Mtg. Invited Session on*

Image Process. Appl. Process Automat., 1991, DSC-vol. 30, pp. 65–62. Also *J. Robotic Systems*, vol. 11, no. 3, pp. 155–168, 1994.

[16] G. V. Puskorius and L. A. Feldkamp, "Global calibration of a robot/vision system," in *Proc. Int. Conf. IEEE Robot. Automat.*, Raleigh, NC, 1987, pp. 190–195.

[17] D. J. Bennett, D. Geiger, and J. M. Hollerbach, "Autonomous calibration for hand-eye coordination," *Int. J. Robot. Res.*, vol. 10, no. 5, pp. 550–559, 1991.

[18] J. Weng, P. Cohen, and M. Herniou, "Camera calibration with distortion models and accuracy evaluation," *IEEE Trans. Patt. Anal. Mach. Intell.*, vol. 14, pp. 965–980, Oct. 1992.

[19] R. K. Lenz and R. Tsai, "Techniques for calibration of the scale factor and image center for high accuracy 3-D machine vision metrology," in *Proc. IEEE Int. Conf. Robot. Automat.*, 1987, pp. 68–75.

[20] A. Bani-Hashemi, "Finding the aspect-ratio of an imaging system," *IEEE Int. Conf. CVPR*, 1991, pp. 122–126.

[21] D. W. Marquardt, "An algorithm for least-squares estimation of nonlinear parameters," *J. Soc. Ind. Appl. Math.*, vol. 11, no. 2, pp. 431–441, June 1963.

[22] H. W. Stone, *Kinematic Modeling, Identification, and Control of Robotic Manipulators.* Boston, MA: Kluwer, 1987.

[23] P. L. Broderick and R. J. Cipra, "A method for determining and correcting robot position and orientation errors due to manufacturing," *ASME J. Mechanisms, Transmissions, and Automation in Design*, vol. 110, no. 3, pp. 3–10, Mar. 1988.

[24] H. Zhuang and Z. S. Roth, "A linear solution to kinematic parameter identification for robot manipulators," *IEEE Trans. Robot. Automat.*, vol. 9, no. 2, pp. 174–185, Apr. 1993.

[25] R. Paul, *Robot Manipulators: Mathematics, Programming, and Control.* Cambridge, MA: MIT Press, 1981.

[26] H. Zhuang, Z. S. Roth, and F. Hamano, "A complete and parametrically continuous kinematic model," *IEEE Trans. Robot. Automat.*, vol. 8, pp. 451–463, Aug. 1992.

[27] H. Zhuang, Z. S. Roth, and K. Wang, "Error-model-based robot calibration using the modified CPC model," *Int. J. Robot. Comp. Integrated Manufact.*, vol. 10, no. 4, pp. 287–299, 1993.

[28] P. E. Gill, W. Murray, and M. H. Wright, *Practical Optimization.* New York: Academic Press, 1981.

[29] T. Kailath, *Linear Systems.* Englewood Cliffs, NJ: Prentice-Hall, 1980.

[30] C. H. Wu, "A kinematic CAD tool for the design and control of a robot manipulator," *Int. J. Robot. Res.*, vol. 3, no. 1, pp. 58–67, 1984.

[31] D. J. Bennett and J. M. Hollerbach, "Autonomous calibration of single-loop closed kinematic chain formed by manipulators with passive endpoint constraints," *IEEE Trans. Robot. Automat.*, vol. 7, pp. 597–606, Oct. 1991.

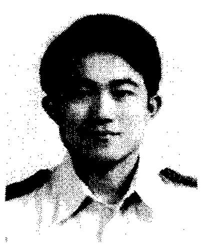
[32] A. Goshtasby, "Correction of image deformation from lens distortion using Bezier patches," *Computer Vision, Graphics, and Image Processing*, vol. 47, pp. 385–394, 1989.

[33] J. H. Borm and C. H. Menq, "Determination of optimal measurement configurations for robot calibration based on observability measure," *Int. J. Robot. Res.*, vol. 10, no. 1, pp. 51–63, Feb. 1991.

[34] M. R. Driels and U. S. Pathre, "Significance of observation strategy on the design of robot calibration experiments," *J. Robot. Syst.*, vol. 7, no. 2, pp. 197–223, 1990.

Hanqi Zhuang (S'88–M'89–SM'93) received the B.S. degree in engineering from Shanghai University of Technology in 1982, and the M.S. and Ph.D. degrees in electrical engineering from Florida Atlantic University, Boca Raton, in 1986 and 1989, respectively.

Since December 1989, he has been a faculty member in the Department of Electrical Engineering, where he is now an Associate Professor. His current research interests include robotics, manufacturing systems, computer vision, and optics.



Kuanchih Wang (S'93–M'93) received the B.S. degree in aeronautical engineering in 1982 from Tamkang University, Taipei, Taiwan, and the M.S. and Ph.D. degrees in electrical engineering in 1990 and 1993 from Florida Atlantic University, Boca Raton.

He is now an Associate Professor in the Electronic Engineering Department of the National I-Lan Institute of Agriculture and Technology, I-Lan City, Taiwan. His research interests include robotics, evolutionary computation, and fuzzy control.

Zvi S. Roth (M'83–SM'92) received the B.S. and M.Sc. degrees in electrical engineering from the Technion, Israel Institute of Technology, Haifa, in 1974 and 1979, respectively, and the Ph.D. degree in systems engineering from Case Western Reserve University, Cleveland, OH, in 1982.

Since then, he has been a faculty member in the Department of Electrical Engineering at Florida Atlantic University, Boca Raton, where he is currently a Professor. He was Director of the FAU Robotics Center from 1985 to 1994. He is now Chairman of the Department of Electrical Engineering. His research interests include estimation, control, and precision measurement aspects of robotic systems. He co-authored the book *Fundamentals of Manipulator Calibration* (Wiley, 1991).